

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Formative Assessment 1

Formative Assessment (FA-1) - Answer Key

Class: Class 9

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Python Fundamentals - Review and Advanced Concepts

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- Answer:** b) [2, 4]
 - Explanation:** x[1:2] starts at index 1 and steps by 2, giving elements at indices 1 and 3: [2, 4].
- Answer:** b) _score
 - Explanation:** Variable names can start with an underscore. They cannot start with a digit (2nd_place), contain hyphens (total-marks), or be reserved keywords (class).
- Answer:** b) <class 'int'>
 - Explanation:** The // operator performs floor division, which returns an integer. $10 // 3 = 3$.
- Answer:** b) *
 - Explanation:** The * operator is used for string repetition (e.g., "ha" * 3 gives "hahaha").
- Answer:** b) False
 - Explanation:** An empty string "" is falsy in Python, so bool("") returns False.

Section II: Short Answer (5 x 2 = 10 marks)

- Differentiate between is and == operators in Python.**
 - Answer:** == checks if two variables have the same **value**, while is checks if they refer to the same **object in memory**.
 - Marking:** 1 mark for explaining == compares values, 1 mark for explaining is compares identity. Example: a = [1,2]; b = [1,2]; a == b is True, but a is b is False.
- Write the output of the swap code.**
 - Answer:** Output: 10 5
 - Explanation:** Python supports simultaneous assignment. a, b = b, a swaps the values of a and b without needing a temporary variable.
 - Marking:** 1 mark for correct output, 1 mark for explanation of tuple unpacking.
- Explain the difference between while loop and for loop.**
 - Answer:** A for loop iterates over a sequence (list, range, string) and is used when the number of iterations is known. A while loop repeats as long as a condition is True and is used when the number of iterations is unknown.
 - Marking:** 1 mark for for loop description, 1 mark for while loop description with use case.
- Write a Python program to check if a number is a palindrome.**
 - Answer:**

```
def is_palindrome(n):  
    original = n  
    reversed_num = 0  
    while n > 0:
```

```

        reversed_num = reversed_num * 10 + n % 10
        n //= 10
    return original == reversed_num

print(is_palindrome(121)) # True
print(is_palindrome(123)) # False

```

- **Marking:** 1 mark for correct logic, 1 mark for working code with test cases.

5. Purpose of break, continue, and pass statements.

- **Answer:**
 - break: Terminates the loop entirely. Example: Exit loop when a condition is met.
 - continue: Skips the current iteration and moves to the next. Example: Skip even numbers in a loop.
 - pass: Does nothing; acts as a placeholder. Example: Used when a statement is syntactically required.
- **Marking:** 1 mark for any two correct explanations with examples, 1 mark for the third.

Section III: Long Answer (5 x 1 = 5 marks)

1. Write a Python program to filter prime numbers from a list.

- **Answer:**

```

def is_prime(n):
    """Check if a number is prime."""
    if n < 2:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

def filter_primes(numbers):
    """Return a list of prime numbers from the input list."""
    return [n for n in numbers if is_prime(n)]

# Test cases
print(filter_primes([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]))
# Output: [2, 3, 5, 7]

print(filter_primes([11, 12, 13, 14, 15]))
# Output: [11, 13]

print(filter_primes([20, 21, 22, 23, 24, 25]))
# Output: [23]

```

- **Marking:** 1 mark for is_prime function, 1 mark for filter_primes function, 1 mark for docstrings, 1 mark for at least 3 test cases, 1 mark for correct output and code style.

Part B: Hands-on Practical Lab Task (5 marks)

- **Answer:**

```

def count_words(sentence):
    return len(sentence.split())

def count_vowels(sentence):
    vowels = "aeiouAEIOU"

```

```

    return sum(1 for ch in sentence if ch in vowels)

def count_consonants(sentence):
    vowels = "aeiouAEIOU"
    return sum(1 for ch in sentence if ch.isalpha() and ch not in vowels)

def count_digits(sentence):
    return sum(1 for ch in sentence if ch.isdigit())

def count_special(sentence):
    return sum(1 for ch in sentence if not ch.isalnum() and ch.isspace() == False)

def vowel_frequency(sentence):
    freq = {}
    for ch in sentence.lower():
        if ch in "aeiou":
            freq[ch] = freq.get(ch, 0) + 1
    return freq

def is_palindrome_sentence(sentence):
    cleaned = ''.join(ch.lower() for ch in sentence if ch.isalnum())
    return cleaned == cleaned[::-1]

# Main
sentence = input("Enter a sentence: ")
print(f"Words: {count_words(sentence)}")
print(f"Vowels: {count_vowels(sentence)}")
print(f"Consonants: {count_consonants(sentence)}")
print(f"Digits: {count_digits(sentence)}")
print(f"Special characters: {count_special(sentence)}")
print(f"Vowel frequency: {vowel_frequency(sentence)}")
print(f"Is palindrome: {is_palindrome_sentence(sentence)}")

```

- **Marking:** 1 mark for functions for each counting operation, 1 mark for vowel frequency, 1 mark for palindrome check ignoring spaces/case, 1 mark for correct logic, 1 mark for clean code structure.

Part C: Notebook Maintenance (5 marks)

1. Neatness and organisation of notebook (2 marks)
2. Completion of all assigned classwork and homework (2 marks)
3. Proper headings, dates, and indexing (1 mark)